MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**
3810 REAGENTS - CAUSTIC SODA&CYANIDE EQP
FDN. DGN. 3810-REA-001,3810-TNK-001&002

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
FDN.	:	Foundation
EQP	:	Equipment
DGN.	:	Design

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1. INTENT

This document is to prepare design of foundation for **3810-TNK-001,002 & 3810-REA-001**.The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2.EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in Excel. For member strength calculation force values are directly taken from Equipment load data drawings & designed in excel sheet.

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4. DESIGN REQUIREMENTS

4.1 Design Load

All Design Loads are indicated in Input drawings attached in Attachment - 1. Also referred Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.2.

5. CONCRETE FOUNDATIONS

5.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304, ACI 318M and API 650.

5.2 Materials

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade

- Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric: ASTM A 185(Plain) / A 497(Deformed)

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5.3 Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Foundation	50
Exposed to weather or contact with ground	Wall	50

5.4 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundation design has been considered 350 Kn/m^2 at 1.2 m below FGL. (Refer Table 5 - BH-15 of Geotechnical Report : MD-216-3000-EG-CV-RPT-1008).

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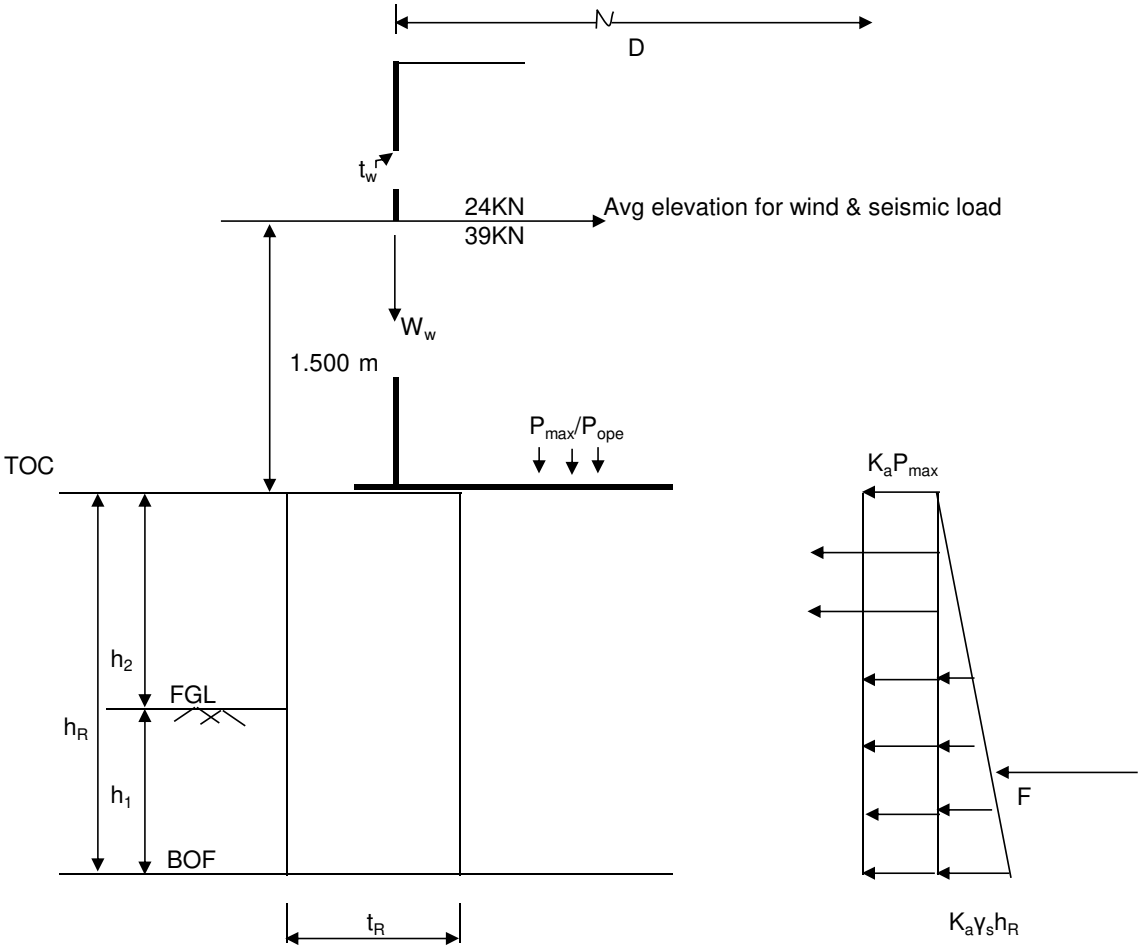
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6. DESIGN CALCULATION FOR FOUNDATION

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Design Calculation of Foundation for 3810-REA-001 & 3810-TNK-001



Typical Section Through Tank

Pressure on ring wall

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Tank Data :

Diameter of tank ,D =	2.90	m
Self weight of tank =	35.0	KN
Self weight of tank per m of circumference, W_w =	3.84	KN/m
Design height of liquid, h_L =	2.7	m
Maximum weight of Fluid=	350	KN
Pressure at base of tank , P_{max}	53	KN/m^2
Operating weight of Fluid=	300	KN
Pressure at base of tank , P_{ope}	45.4	KN/m^2

Foundation Material :

Concrete, f_c' =	35	Mpa
Reinforcement, f_y =	420	Mpa
Unit weight of concrete, γ_c =	24	Kn/m^3

Soil Data :

Unit weight of soil, γ_s =	20	Kn/m^3
Allowable bearing pressure, q_{all}=	350	Kn/m^2
Coefficient of active earth pressure, K_a =	0.33	

Design Data of Ringwall :

Ringwall depth, $h_R = h_1 + h_2$ =	1.9	m
FGL to Bottom of foundation , h_1 =	1.2	m
FGL to Top of concrete , h_2 =	0.7	m
Ringwall thickness, t_R =	500	mm
Inner dia of ringwall, D_i =	2.2	m
Outer dia of ringwall, D_o =	3.2	m
$S_{xx} = \pi(D_o^4-D_i^4)/32D_o$ =	2.50	m^3

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Check of soil bearing pressure (Max Fluid Load): For wind Load

Wind force = 39 kn acting at a height of 3.40 m from BOF

M_{OT} = Overturning moment = 132.6 KN-m

M_{OT} / S_{xx} = 53.1 KN/m^2

Load Combination : Dead + Fluid

$q_{RF} = W_w/t_R + P_{max}/2 + h_R\gamma_c$ = 79.8 KN/m^2

Load Combination : Dead + Fluid +/- Wind

$q_{max} = q_{RF} + M_{OT}/S_{xx}$ = 133 KN/m^2 < 350.0 Kn/m^2 OK

$q_{min} = q_{RF} - M_{OT}/S_{xx}$ = 26.7 KN/m^2

Check of soil bearing pressure (Empty Tank): For wind Load

Soil bearing pressure under ringwall :

Load Combination : Dead +/- Wind

$q_{RE} = W_w/t_R + h_R\gamma_c$ = 53.3 KN/m^2

$q_{max} = q_{RE} + M_{OT}/S_{xx}$ = 106.4 KN/m^2

$q_{min} = q_{RE} - M_{OT}/S_{xx}$ = 0.21 KN/m^2 no -ve pressure

Check for overturning (Empty Tank): For wind Load

M_{OT} = Overturning moment = 132.6 KN-m

M_{OR} = Restoring moment = (35+PI()*2.9*0.5*1.9*25)*2.9/2 = 364.5 KN-m

FOS = M_{OR} / M_{OT} 2.75 > 1.5 ok

Check for soil bearing pressure (Operating Fluid Load): For Seismic Load

Seismic force at center of mass of equipment = 24 KN

acting at a height of 3.40 m from BOF

M_{OT} = Overturning moment = 81.6 KN-m

M_{OT} / S_{xx} = 32.7 KN/m^2

Load Combination : Dead + Fluid

$q_{RF} = W_w /t_R + P_{ope}/2 + h_R\gamma_c$ = 76 KN/m^2

Load Combination : Dead + Fluid +/- 0.7Earthquake load

$q_{max} = q_{RF} + 0.7M_{OT}/S_{xx}$ = 98.9 KN/m^2 < 350.0 Kn/m^2 OK

$q_{min} = q_{RF} - 0.7M_{OT}/S_{xx}$ = 53.1 KN/m^2

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Check for differential settlement (Max Slurry Load): For Static condition

Soil bearing pressure under ringwall :

$q_{RF} = W_w/t_R + 0.7P_{max} + h_R\gamma_c$ = 90.4 Kn/m^2 < 350.0 KN/m^2 Ok

Soil bearing pressure under tank :

$q_{TF} = \gamma_p h_L + h_R\gamma_s$ = 91.0 Kn/m^2 < 350.0 KN/m^2 Ok

As the tank encroaches the ring wall by 350mm out of total 500mm , the pressure under the tank is reduced to 350/500 = 0.7 times.

Soil bearing pressure under ring wall and tank bottom is almost equal, therefore, no settlement problem will occur.

Ringwall Reinforcement :

Radial outward force on ringwall, $F = K_a h_R(P_{max} + \gamma_s h_R/2)$ = 45.1 Kn/m

For Hydrotest :

Radial outward force on ringwall, $F = K_a h_R(\gamma_w h_L + \gamma_s h_R)$ = 28.8 Kn/m

Hoop tension in ringwall, T = 1/2FD = 65.4 Kn

Factored design tension, T_u = 91.6 Kn FOS = 1.4

$A_{s,reqd} = T_u/\phi f_y$ = 242 mm^2 ϕ = 0.9

$A_{s,min} = 0.0025 h_R t_R$ = 1900 mm^2

$A_{s,reqd}$ (per face of ring wall)= 950 mm^2

Required : 12 dia bar @ 226 c/c (per face)

Provide : 12 dia bar @ 150 c/c (per face)

$A_{s,\ provided}$(per face) = 1433 mm^2 ok

Vertical Reinforcement :

$A_{s,min} = 0.0015 t_R$ = 600 mm^2/m

$A_{s,reqd}$ (per face of ring wall)= 300 mm^2

Required : 10 dia bar @ 262 c/c (per face)

Provide : 10 dia bar @ 200 c/c (per face)

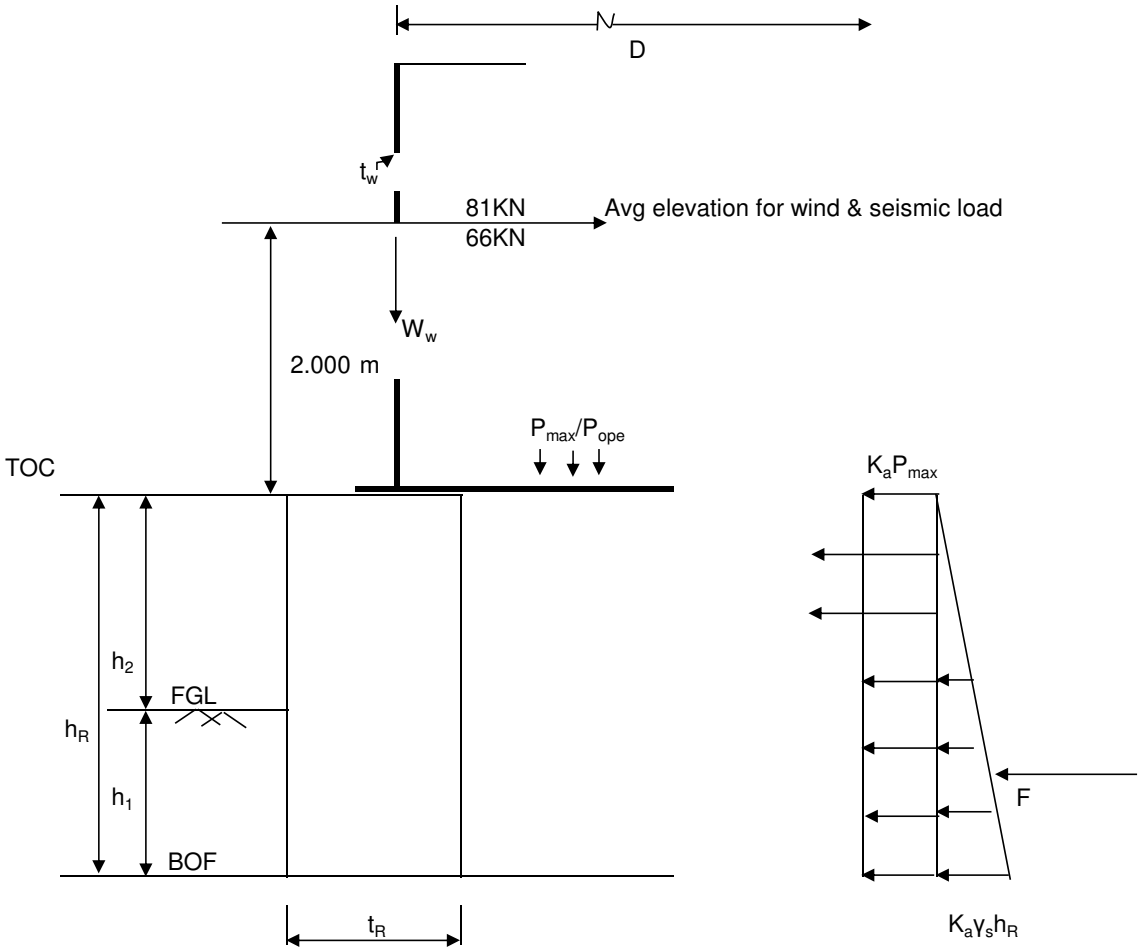
$A_{s,\ provided}$(per face) = 393 mm^2 ok

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Design Calculation of Foundation for 3810-TNK-002



Typical Section Through Tank

Pressure on ring wall

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Tank Data :

Diameter of tank ,D =	4.00	m
Self weight of tank =	85.0	KN
Self weight of tank per m of circumference, W_w =	6.76	KN/m
Design height of liquid, h_L =	3.7	m
Maximum weight of Fluid=	760	KN
Pressure at base of tank , P_{max}	60.5	KN/m^2
Operating weight of Fluid=	600	KN
Pressure at base of tank , P_{ope}	47.7	KN/m^2

Foundation Material :

Concrete, f_c' =	35	Mpa
Reinforcement, f_y =	420	Mpa
Unit weight of concrete, γ_c =	24	Kn/m^3

Soil Data :

Unit weight of soil, γ_s =	20	Kn/m^3
Allowable bearing pressure, q_{all}=	350	Kn/m^2
Coefficient of active earth pressure, K_a =	0.33	

Design Data of Ringwall :

Ringwall depth, $h_R = h_1 + h_2$ =	1.9	m
FGL to Bottom of foundation , h_1 =	1.2	m
FGL to Top of concrete , h_2 =	0.7	m
Ringwall thickness, t_R =	500	mm
Inner dia of ringwall, D_i =	3.25	m
Outer dia of ringwall, D_o =	4.25	m
$S_{xx} = \pi(D_o^4-D_i^4)/32D_o$ =	4.96	m^3

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Check of soil bearing pressure (Max Fluid Load): For wind Load

Wind force = 66 kn acting at a height of 3.90 m from BOF

M_{OT} = Overturning moment = 257.4 KN-m

M_{OT} / S_{xx} = 51.9 KN/m^2

Load Combination : Dead + Fluid

$q_{RF} = W_w/t_R + P_{max}/2 + h_R\gamma_c$ = 89.4 KN/m^2

Load Combination : Dead + Fluid +/- Wind

$q_{max} = q_{RF} + M_{OT}/S_{xx}$ = 141 KN/m^2 < 350.0 Kn/m^2 OK

$q_{min} = q_{RF} - M_{OT}/S_{xx}$ = 37.5 KN/m^2

Check of soil bearing pressure (Empty Tank): For wind Load

Soil bearing pressure under ringwall :

Load Combination : Dead +/- Wind

$q_{RE} = W_w/t_R + h_R\gamma_c$ = 59.1 KN/m^2

$q_{max} = q_{RE} + M_{OT}/S_{xx}$ = 111.0 KN/m^2

$q_{min} = q_{RE} - M_{OT}/S_{xx}$ = 7.23 KN/m^2 no -ve pressure

Check for overturning (Empty Tank): For wind Load

M_{OT} = Overturning moment = 257.4 KN-m

M_{OR} = Restoring moment = (85+PI()*4*0.5*1.9*25)*4/2 = 766.9 KN-m

FOS = M_{OR} / M_{OT} 2.98 > 1.5 ok

Check for soil bearing pressure (Operating Fluid Load): For Seismic Load

Seismic force at center of mass of equipment = 81 KN

acting at a height of 3.90 m from BOF

M_{OT} = Overturning moment = 316 KN-m

M_{OT} / S_{xx} = 63.7 KN/m^2

Load Combination : Dead + Fluid

$q_{RF} = W_w /t_R + P_{ope}/2 + h_R\gamma_c$ = 83 KN/m^2

Load Combination : Dead + Fluid +/- 0.7Earthquake load

$q_{max} = q_{RF} + 0.7M_{OT}/S_{xx}$ = 128 KN/m^2 < 350.0 Kn/m^2 OK

$q_{min} = q_{RF} - 0.7M_{OT}/S_{xx}$ = 38.4 KN/m^2

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Check for differential settlement (Max Slurry Load): For Static condition

Soil bearing pressure under ringwall :

$q_{RF} = W_w/t_R + 0.65P_{max} + h_R\gamma_c$ = 98.4 Kn/m^2 < 350.0 KN/m^2 Ok

Soil bearing pressure under tank :

$q_{TF} = \gamma_p h_L + h_R\gamma_s$ = 98.5 Kn/m^2 < 350.0 KN/m^2 Ok

As the tank encroaches the ring wall by 325mm out of total 500mm , the pressure under the tank is reduced to 325/500 = 0.65 times.

Soil bearing pressure under ring wall and tank bottom is almost equal, therefore, no settlement problem will occur.

Ringwall Reinforcement :

Radial outward force on ringwall, $F = K_a h_R(P_{max} + \gamma_s h_R/2)$ = 49.8 Kn/m

For Hydrotest :

Radial outward force on ringwall, F= $K_a h_R(\gamma_w h_L + \gamma_s h_R)$ = 35.1 Kn/m

Hoop tension in ringwall, T = 1/2FD = 99.7 Kn

Factored design tension, T_u= 140 Kn FOS = 1.4

$A_{s,reqd} = T_u/\phi f_y$ = 369 mm^2 ϕ = 0.9

$A_{s,min} = 0.0025 h_R t_R$ = 1900 mm^2

$A_{s,reqd}$ (per face of ring wall)= 950 mm^2

Required : 12 dia bar @ 226 c/c (per face)

Provide : 12 dia bar @ 150 c/c (per face)

$A_{s,\ provided}$(per face) = 1433 mm^2 ok

Vertical Reinforcement :

$A_{s,min} = 0.0015 t_R$ = 600 mm^2/m

$A_{s,reqd}$ (per face of ring wall)= 300 mm^2

Required : 10 dia bar @ 262 c/c (per face)

Provide : 10 dia bar @ 200 c/c (per face)

$A_{s,\ provided}$(per face) = 393 mm^2 ok

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REFERENCES

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

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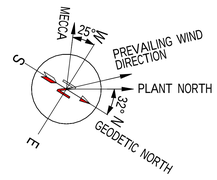
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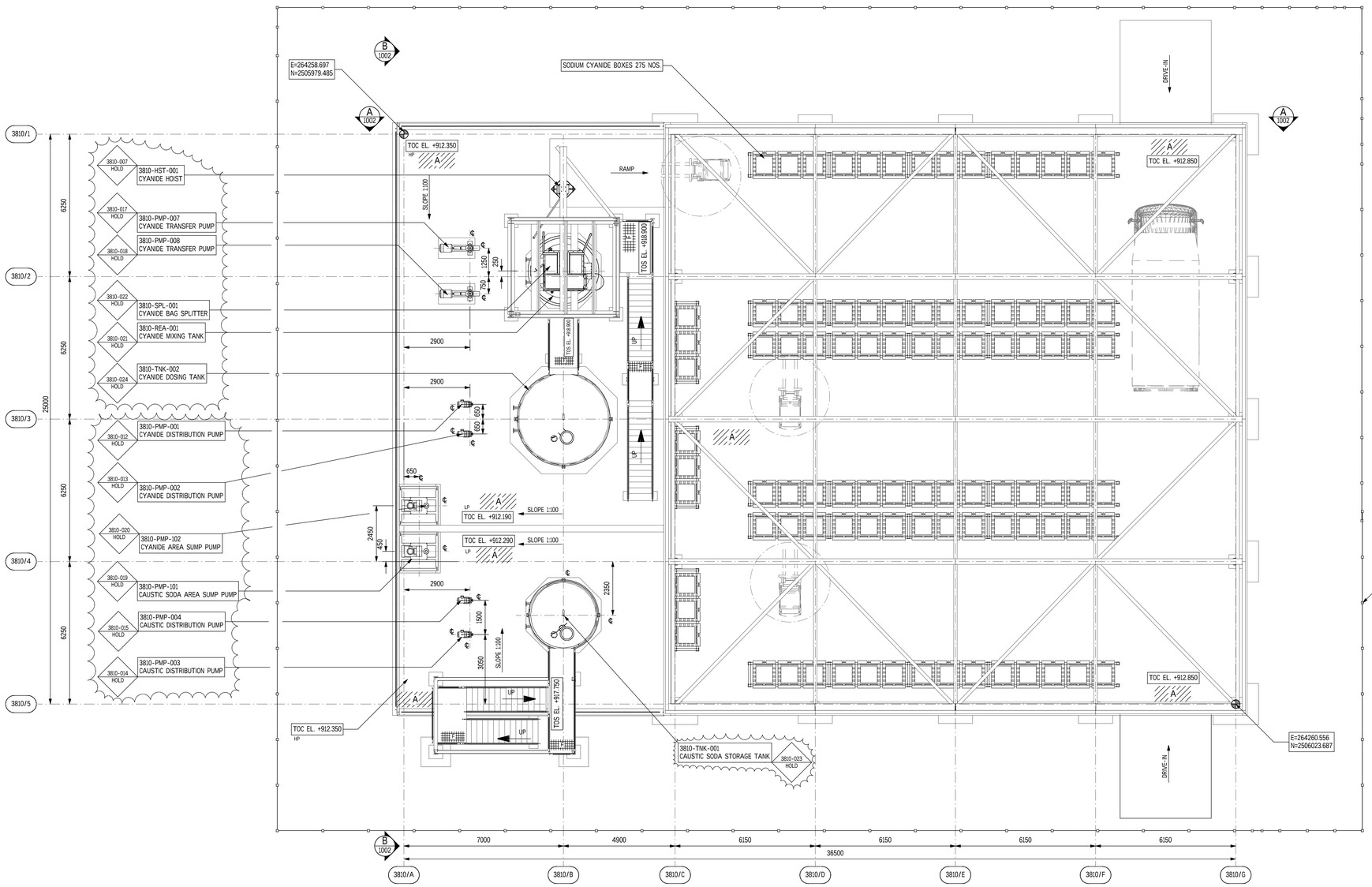
ATTACHMENT 1 INPUT DRAWINGS

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DRAWING NUMBER: MD-216-3810-EG-ME-GAD-1001 REV A02



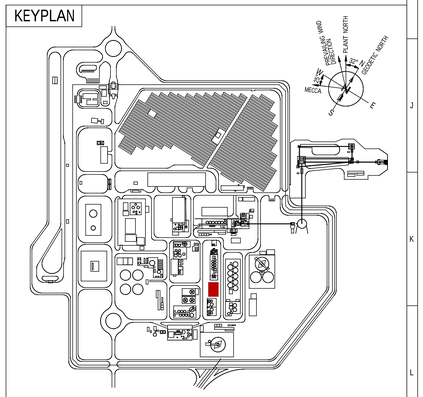


PLAN VIEW

- CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
- CONCRETE FLOOR
- CONCRETE FLOOR WITH ALKALI/ ACID RESISTANT COATING
- CHECKERED PLATE
- GRATING
- GRATING - FRP
- ACID RESISTANT BRICKLINING
- BRICK WALL
- CONCRETE WALL

1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METERS UNLESS OTHERWISE STATED.
2. FINISHED GROUND LEVEL (F.G.L) FOR PROCESS AREA 3810 = RL(+)912.350 m
3. F.F.L, F.G.L, T.O.C = FINISHED FLOOR LEVEL, FINISHED GROUND LEVEL, TOP OF CONCRETE. T.O.S, B.O.S, B.O.B, = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE. T.O.G. = TOP OF GRATING. HP = HIGHT POINT. LP = LOW POINT.
4. BUILDING DIMENSIONS AND HEIGHTS ARE TENTATIVE, SHALL BE FINALIZED DURING DETAIL ENGINEERING SUBJECT TO MINIMUM AS SHOWN.
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



0 1 10m

DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3000-EG-GM-LST-1010	HOLD LIST
MD-216-3810-EG-ME-GAD-1002	SECTION A-A & B-B

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A02	16-01-2020	ISSUED FOR DESIGN	PP	TV	SC	RL	
A01	15-10-2019	ISSUED FOR APPROVAL	AM	TV	SC	RL	

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LARSEN & TOUBRO DRAWING NUMBER: 019045_450_M_G_03251_001 — REVISION: 1

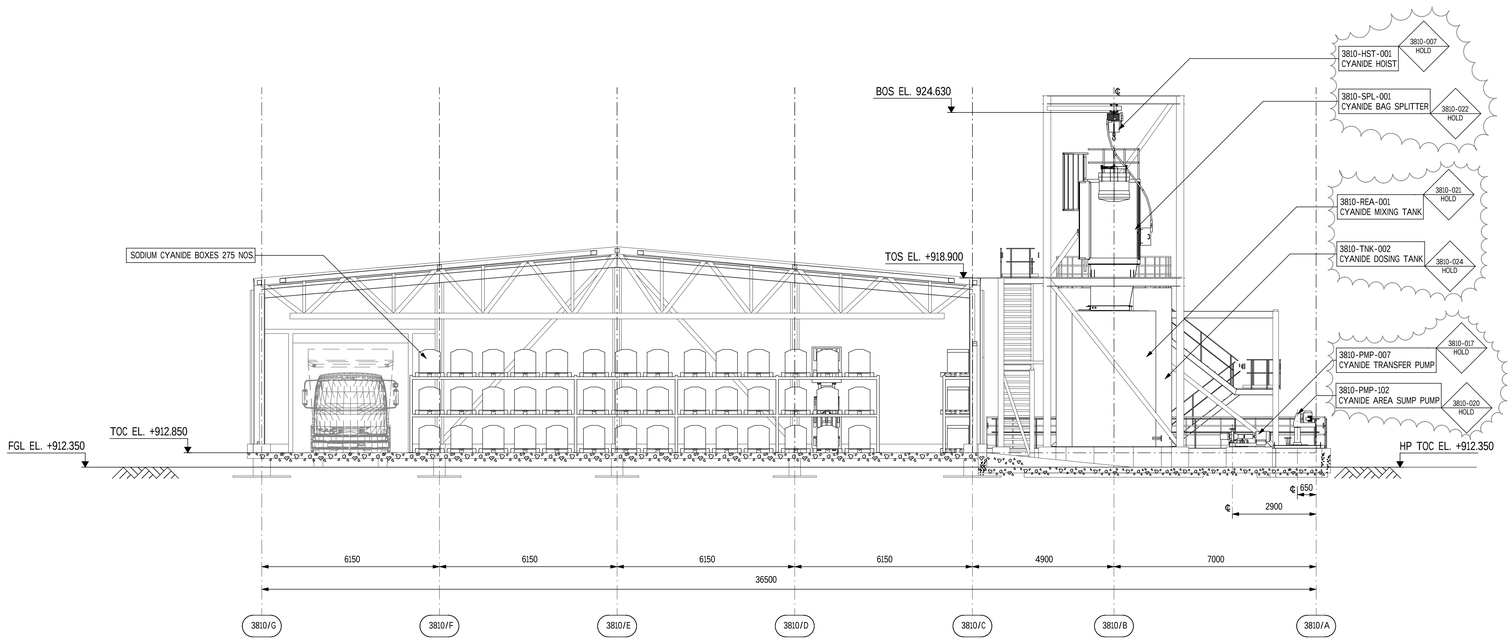
OUTOTEC DRAWING NUMBER: 900707 _GAU01 _AA24_ MTL01 _ 38101 — DOCUMENT ID: OU100664255 — REVISION: 1

DRAWN	AM	15-10-2019
CHECKED	TV	15-10-2019
ENGINEER	SC	15-10-2019
APPROVED	RL	15-10-2019
EMPLOYER		
SCALE	1:75	A0

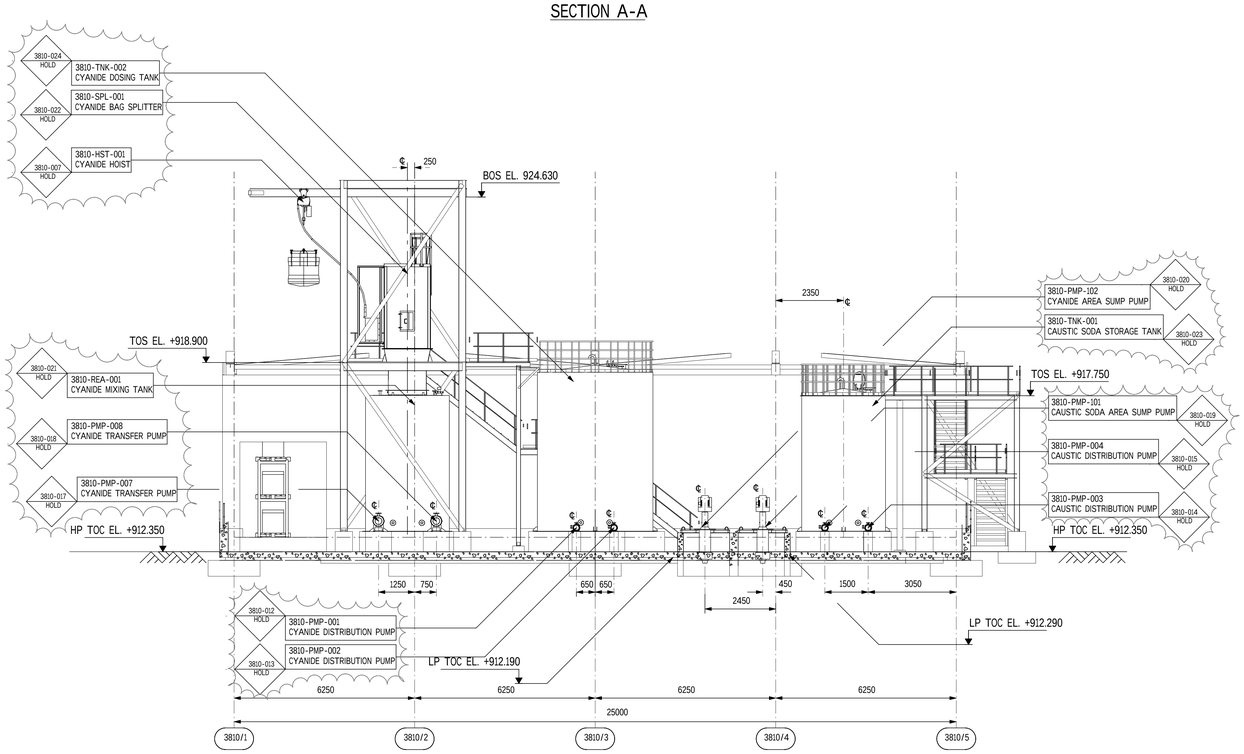
MANSOURAH-MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
REAGENTS - CAUSTIC AND CYANIDE
PLAN VIEW

DRAWING NUMBER: MD-216-3810-EG-ME-GAD-1001 — REV: A02

DRAWING NUMBER: MD-216-3810-EG-ME-GAD-1002 REV: A02



SECTION A-A

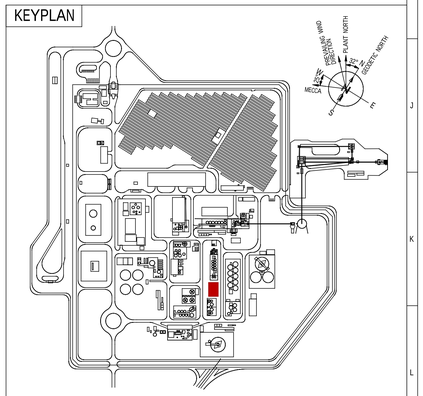


SECTION B-B

- CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
- CONCRETE FLOOR
- A — CONCRETE FLOOR WITH ALKALI/ ACID RESISTANT COATING
- CHECKERED PLATE
- GRATING
- F — GRATING - FRP
- ACID RESISTANT BRICKLINING
- BRICK WALL
- CONCRETE WALL

1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METERS UNLESS OTHERWISE STATED.
2. FINISHED GROUND LEVEL (F.G.L) FOR PROCESS AREA 3810 = RL(+)912.350 m
3. F.F.L, F.G.L, T.O.C = FINISHED FLOOR LEVEL, FINISHED GROUND LEVEL, TOP OF CONCRETE. T.O.S, B.O.S, B.O.B, = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE. T.O.G. = TOP OF GRATING. HP = HIGHT POINT. LP = LOW POINT.
4. BUILDING DIMENSIONS AND HEIGHTS ARE TENTATIVE, SHALL BE FINALIZED DURING DETAIL ENGINEERING SUBJECT TO MINIMUM AS SHOWN.
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



0 1 10m

DOCUMENT NUMBER	REFERENCE DOCUMENTS	DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3000-EG-GM-LST-1010	HOLD LIST		
MD-216-3810-EG-ME-GAD-1001	PLAN VIEW		

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A02	16-01-2020	ISSUED FOR DESIGN	PP	TV	SC	RL	
A01	15-10-2019	ISSUED FOR APPROVAL	AM	TV	SC	RL	

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

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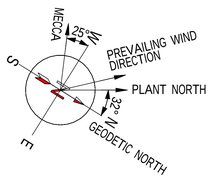
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LARSEN & TOUBRO DRAWING NUMBER	019045_450_M_G_03251_002		REVISION 1
OUTOTEC DRAWING NUMBER	900707 _GAU01 _AA24_MTL01 _38102	DOCUMENT ID OU100664256	REVISION 1
DRAWN	AM	15-10-2019	MANSOURAH-MASSARAH GOLD PROJECT
CHECKED	TV	15-10-2019	GENERAL ARRANGEMENT DRAWING
ENGINEER	SC	15-10-2019	REAGENTS - CAUSTIC AND CYANIDE
APPROVED	RL	15-10-2019	SECTION A-A & B-B
EMPLOYER			DRAWING NUMBER: MD-216-3810-EG-ME-GAD-1002 REV: A02
SCALE	1:75	A0	

DRAWING NUMBER: MD-216-3810-EG-ME-GAD-1011 | REV: A01



Cyanide code compliance is the minimum standard. Security fencing, cameras, % volume bunding are key drivers.

E=264258.697
N=2505979.485

ensure this area meets ICMC requirements for containment and bunding

Consider this area has totally different management from security, warehouse and operation as we discussed before. On the other hand Caustic Soda has another operation of offloading from trucks and so on. So consider to separate cyanide area by fence from Caustic Soda

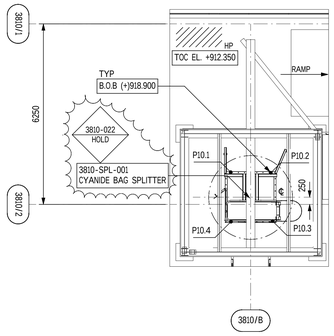
3810-017 HOLD — 3810-PMP-007 CYANIDE TRANSFER PUMP
3810-018 HOLD — 3810-PMP-008 CYANIDE TRANSFER PUMP
3810-021 HOLD — 3810-REA-001 CYANIDE MIXING TANK
3810-024 HOLD — 3810-TNK-002 CYANIDE DOSING TANK
3810-012 HOLD — 3810-PMP-001 CYANIDE DISTRIBUTION PUMP
3810-013 HOLD — 3810-PMP-002 CYANIDE DISTRIBUTION PUMP
3810-020 HOLD — 3810-PMP-102 CYANIDE AREA SUMP PUMP
3810-019 HOLD — 3810-PMP-101 CAUSTIC SODA AREA SUMP PUMP
3810-015 HOLD — 3810-PMP-004 CAUSTIC DISTRIBUTION PUMP
3810-014 HOLD — 3810-PMP-003 CAUSTIC DISTRIBUTION PUMP
3810-TNK-001 CAUSTIC SODA STORAGE TANK — 3810-023 HOLD

TOC EL. +912.350 HP
TOC EL. +912.850
TOC EL. +912.190
TOC EL. +912.290
TOS EL. +918.900
TOS EL. +918.300
TOS EL. +917.750
RAMP
DRIVE-IN
UP
SLOPE 1:100
TYP
B.O.B (+)913.050
B.O.B (+)913.000
(ID OF TANK) ⌀ 2900
(ID OF TANK) ⌀ 4000
⌀ 2900 (ID OF TANK)
300 x 260 TYP
4 AXLE RIGID TRUCK FRONT AXLE LOAD 256KN
TOPIK LIFT AXLE LOAD 68KN
FENCING
E=264260.556
N=2506023.687

P1 P2 P3 P4 P5 P6 P7 P8 P9 P11 P12

3810/1 3810/2 3810/3 3810/4 3810/5
3810/A 3810/B 3810/C 3810/D 3810/E 3810/F 3810/G
6250 6250 6250 6250 25000
7000 4900 6150 6150 6150 6150
36500

PLAN VIEW



PLAN AT EL+ 918.900

LOADING POINT	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10.1/...P10.4	P11	P12
LOAD CASE	Vz(kN) hx(kN) hy(kN)	Vz(kN) hx(kN) hy(kN)	Vz(kN) hx(kN) hy(kN)	Vz(kN) hx(kN) hy(kN)	Vz(kN) hx(kN) hy(kN)	Vz(kN) hx(kN) hy(kN)	Vz(kN) hx(kN) hy(kN)	Vz(kN) hx(kN) hy(kN)	Vz(kN) hx(kN) hy(kN)	Vz(kN) hx(kN) hy(kN)	Vz(kN) hx(kN) hy(kN)	Vz(kN) hx(kN) hy(kN)
1 DEAD LOAD EQUIPMENT	- 3	- 3	- 3	- 3	- 35	- 85	- 7	- 7	- 35	- 4	- 4	- 4
2 DEAD LOAD (SECONDARY EQUIPMENT)												
3 OPERATION LOAD 1. FILLING					- 300	- 600			- 300	- 6		
4 OPERATION LOAD 2. PRESSURE												
5 OPERATION LOAD 3. TEMPERATURE												
6 OPERATION LOAD 4. DYNAMIC	- 5	- 5	- 5	- 5			- 10	- 10			- 6	- 6
7 OPERATION LOAD 5. OTHERS												
8 TEST LOAD FILLING / EMERGENCY												
9 LIVE LOAD					- 350	- 760			- 350			
10 WIND ±X - DIRECTION	- - -	- - -	- - -	- - -	- ± 39 ↑ 39	- ± 66 ↑ 66	- - -	- - -	- ± 39 ↑ 39	- - -	- - -	- - -
11 WIND ±Y - DIRECTION	- - -	- - -	- - -	- - -	- ± 39 ↑ 39	- ± 66 ↑ 66	- - -	- - -	- ± 39 ↑ 39	- - -	- - -	- - -
12 EARTHQUAKE ±X - DIRECTION	± 1 ± 1	± 1 ± 1	± 1 ± 1	± 1 ± 1	± 24 ↑ 24	± 61 ↑ 61	± 1 ± 1	± 1 ± 1	± 24 ↑ 24	± 1 ± 1	± 1 ± 1	± 1 ± 1
13 EARTHQUAKE ±Y - DIRECTION	± 1 ± 1	± 1 ± 1	± 1 ± 1	± 1 ± 1	± 24 ↑ 24	± 61 ↑ 61	± 1 ± 1	± 1 ± 1	± 24 ↑ 24	± 1 ± 1	± 1 ± 1	± 1 ± 1
EQUIPMENT - NO.	3810-PMP-001	3810-PMP-002	3810-PMP-003	3810-PMP-004	3810-TNK-001	3810-TNK-002	3810-PMP-101	3810-PMP-102	3810-REA-001	3810-SPL-001	3810-PMP-007	3810-PMP-008
DESCRIPTION	CYANIDE DIST.PUMP	CYANIDE DIST.PUMP	CAUSTIC DIST.PUMP	CAUSTIC DIST.PUMP	CAUSTIC TANK	CYANIDE TANK	CAUSTIC SUMP PUMP	CYANIDE SUMP PUMP	CYANIDE TANK	CYANIDE SPLITTER	CYANIDE PUMP	CYANIDE PUMP

- CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
- CONCRETE FLOOR
- CONCRETE FLOOR WITH ALKALI/ ACID RESISTANT COATING
- CHECKERED PLATE
- GRATING
- GRATING - FRP
- ACID RESISTANT BRICKLINING
- BRICK WALL
- CONCRETE WALL

GENERAL NOTES, LEGEND

01 - ALL DIMENSION ARE IN MILLIMETER. 02 - ALL COORDINATE AND LEVELS ARE IN METER.

03 - LOADS ARE INDICATED IN KILONEWTON[kN]; KILONEWTONMETER[kNm] OR KILONEWTON/ SQUAREMETER [kN/m²] UNLESS OTHERWISE NOTED.

04 - ALL LOADS REFER TO TOP OF BEAM UNLESS OTHERWISE NOTED.

05 - T.O.S., B.O.S, B.O.B. = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
T.O.C. = TOP OF CONCRETE
LP = LOW POINT.
HP = HIGHT POINT.

DIRECTION OF FORCES AND MOMENTS: +Z +Mz +Y +X +Mx +My

06 - THE LOADS SHOWN DO NOT INCLUDE :
- LOADS FROM AIR CONDITIONING EQUIPMENT
- ADDITIONAL LOADS FOR ERECTION PURPOSES

07 - FLOOR DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:

OPERATING PLATFORM- GRATING 5 kN/m² — GROUND FLOOR NOT ACCESSIBLE TO TRUCKS 10 kN/m²

HEAVY STORAGE AREAS ACCESSIBLE BY VEHICLE OR CRANE 30 kN/m²
ELECTRICAL SWITCH GEAR AREAS 10 kN/m²

08 - ROOF DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:

DUST 0.5 kN/m² ROOF NOT ACCESSED FOR MAINTENANCE 0.8kN POINT LOAD

09 - FOR ITEM SUCH AS DUCTS, PIPES, CABLE TRAYS, ETC. NOT SHOWN SEPARATELY, AN ADDITIONAL VERTICAL LOAD OF 50 kN SHALL BE PROVIDED ON TOP OF EACH COLUMN.

10 - STATIC LOADS :
FOR STATIC LOADS CERTAIN EQUIPMENT DATA ARE TAKEN INTO CONSIDERATION, AFTER PLACEMENT OF PURCHASE ORDER, STATIC LOADS FOR CERTAIN EQUIPMENT HAVE TO BE RECHECKED ACCORDING TO ACTUAL WEIGHTS.

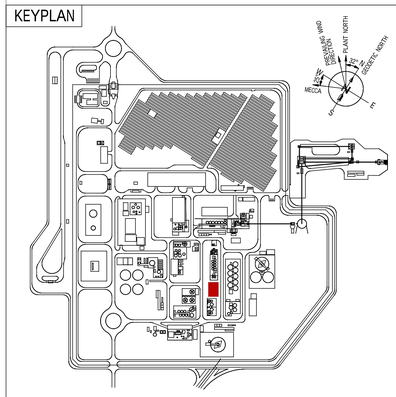
11 - DYNAMIC LOADS :
INFORMATION WITH REGARD TO DYNAMIC LOADS WHICH ARE NOT INDICATED AND HAVE TO BE REQUESTED WITH THE SUPPLIER OF THE RESPECTIVE EQUIPMENT, THIS INFORMATION IS REQUIRED FOR THE DETAIL ENGINEERING AS A ROUGH ESTIMATION FOR THE DYNAMIC LOADS. TAKE 1.5 TIMES OF THE STATIC LOADS , BASED ON THE INDICATIONS FOR USUAL EQUIPMENT SUPPLIERS.

12 - FINISHED GROUND LEVEL (F.G.L.) = RL(+)912.350 m.

13 - LOCATIONS AND SIZES OF PUMP FOUNDATION ARE TENTATIVE AND WILL BE SHOWN IN DETAIL DESIGN PHASE.

14 - ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



0 1 10m

BECHTEL

Document Number
MD-216-3810-EG-ME-GAD-1011

Revision A01

Approved with Comments

Mr Doron Golan - Bechtel Global

Feb 13, 2020, 10:06 PM GMT+2:00

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DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3000-EG-GM-LST-1010	HOLD LIST
MD-216-3810-EG-ME-GAD-1001	GENERAL ARRANGEMENT, PLAN VIEW

DOCUMENT NUMBER	REFERENCE DOCUMENTS

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A01	23-01-2020	ISSUED FOR APPROVAL	PP	TV	SC	RL	

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

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LARSEN & TOUBRO DRAWING NUMBER	REVISION
019045_450_M_D_03259_001	0
OUTOTEC DRAWING NUMBER: 900707 _GAU01 _AA24_ MLD14 _38101	DOCUMENT ID: OU100667839 — REVISION: 0

DRAWN	PP	23-01-2020
CHECKED	TV	23-01-2020
ENGINEER	SC	23-01-2020
APPROVED	RL	23-01-2020
EMPLOYER		
SCALE	1:75	A0

MANSOURAH-MASSARAH GOLD PROJECT
LOAD PLAN
REAGENTS - CAUSTIC AND CYANIDE
PLAN VIEW

DRAWING NUMBER: MD-216-3810-EG-ME-GAD-1011 | REV: A01